

1.) $\Rightarrow$: Fix $\varepsilon > 0$. As $\lim a_n = \infty$, then $\exists N \in \mathbb{N}$ s.t.

$$a_n > \frac{1}{\varepsilon} \quad \forall n \geq N.$$

So, for $n \geq N$,

$$|\frac{1}{a_n} - 0| = \frac{1}{a_n} < \varepsilon$$

Therefore $\lim \frac{1}{a_n} = 0$.

$\Leftarrow$: Fix $M > 0$. As $\lim \frac{1}{a_n} = 0$, then $\exists N \in \mathbb{N}$ s.t.

$$0 < \frac{1}{a_n} = |\frac{1}{a_n} - 0| < \frac{1}{M} \quad \forall n \geq N$$

So, for $n \geq N$,

$$a_n > M$$

Therefore $\lim a_n = \infty$. □

2.) Fix $\varepsilon > 0$. As $\sum |a_n|$ converges, then by the Cauchy criterion, $\exists N \in \mathbb{N}$ s.t.

$$\sum_{k=m}^n |a_k| < \varepsilon \quad \forall n \geq m \geq N.$$

So, for $n \geq m \geq N$, we have

$$\left| \sum_{k=m}^n a_{2k} \right| \leq \sum_{k=m}^n |a_{2k}| \leq \sum_{l=2m}^{2n} |a_l| < \varepsilon$$

↑
since $2n \geq 2m \geq 2N \geq N$.

Therefore $\sum a_{2n}$ converges by the Cauchy criterion. □

3.) $\Rightarrow$: Suppose $A \subseteq X$ is open. Then $\forall a \in A$

$\exists r_a > 0$ s.t. $B_{r_a}(a) \subseteq A$. We claim that

$$A = \bigcup_{a \in A} B_{r_a}(a),$$

which will show that $I=A$ is a set that works and so finish the proof. Note that

$$x \in A \Rightarrow x \in B_{r_x}(x) \subseteq A \Rightarrow x \in \bigcup_{a \in A} B_{r_a}(a),$$

and so $A \subseteq \bigcup_{a \in A} B_{r_a}(a)$. Conversely,

$$\begin{aligned} y \in \bigcup_{a \in A} B_{r_a}(a) &\Rightarrow \exists a \in A \text{ s.t. } y \in B_{r_a}(a) \\ &\Rightarrow y \in B_{r_a}(a) \subseteq A \Rightarrow y \in A, \end{aligned}$$

and so $A \supseteq \bigcup_{a \in A} B_{r_a}(a)$. Together, we conclude

$$A = \bigcup_{a \in A} B_{r_a}(a), \text{ as desired.}$$

$\Leftarrow$: Suppose $A = \bigcup_{i \in I} B_{r_i}(a_i)$ for some set I , numbers $r_i > 0$, and points $a_i \in X$. From lecture, we know $B_{r_i}(a_i)$ is open $\forall i \in I$. So A is open, since the arbitrary union of open sets is open. $\square$

4.) (a.) Prove: By lecture, it suffices to show that Y is open in $(\mathbb{R}^2, d_2)$. Fix $(x, y) \in Y$. Then $y > 0$, and $B_y((x, y)) \subseteq Y$ since

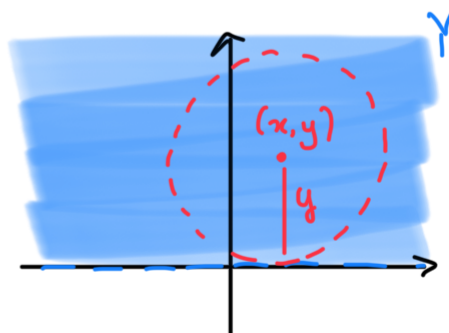
$$(x', y') \in B_y(x, y) \Rightarrow \sqrt{|x-x'|^2 + |y-y'|^2} < y$$

$$\Rightarrow |y-y'| \leq \sqrt{|x-x'|^2 + |y-y'|^2} < y$$

$$\Rightarrow y' > y - (y-y') > y - y = 0$$

$$\Rightarrow (x', y') \in Y.$$

So Y is open.



(b.) Disprove: Consider $B=Y$. Then

- Y is closed in (Y, d_Y) (this is true for any metric space)

- Y is not closed in $(\mathbb{R}^2, d_2)$: Note that $(0,0) \in \overline{Y}^{\mathbb{R}^2}$, since given $r > 0$, the point $(0, \frac{r}{2})$ is in Y and $B_r^{\mathbb{R}^2}((0,0))$. So

$\overline{Y}^{\mathbb{R}^2} \neq Y$, and thus Y is not closed in $(\mathbb{R}^2, d_2)$.

□

